Analysis and Design of a Generalized Class E Tuned Power Amplifier

CHARALAMPOS P. AVRATOGLOU, NICHOLAS C. VOULGARIS, AND FOTINI I. IOANNIDOU

Abstract — **The analysis of a more general Class E amplifier configuration is described which can yield results for most of the other class E amplifier configurations published thus far, by means of simply imposing the proper design conditions. Furthermore, this analysis takes into account most of the important parameters which affect the amplifier performance, such as the dc-feed inductance, the Q_L factor of the series-tuned circuit and the switching device ON resistance. Design and performance curves are given and discussed for the finite dc-feed inductance, shunt-capacitor, series-tuned configuration. It is seen that the amplifier performance is strongly affected by all of these parameters. Laboratory results show an excellent agreement between experimental values and theoretical predictions.**

I. Introduction

THE Class E amplifier [1] configuration with finite dc-feed inductance, shown in Fig. 1, may be described as a generalized Class E amplifier configuration, since its analysis can provide results for most of the other Class E amplifier configurations published up to the present time. This can be done by reducing to zero the values of the components which do not exist in the other circuit configurations and by imposing the proper design conditions to the analysis equations.

Such a configuration has already been analyzed by Zulinski [2] using a method similar to that given by Raab [3] which does not take into account a) the Q_L factor of the series-tuned circuit restricting thus the applicability of the results to the high-Q_L region, and b) the active device ON resistance.

In this paper a complete analysis of the above-mentioned generalized configuration is carried out using the analysis method proposed in [4]. The analysis takes into account all the important parameters which affect the amplifier performance, i.e., the dc-feed inductance, the Q_L factor of the series-tuned circuit and the switching device ON resistance. Thus the analysis can provide very useful and accurate design and performance data not available before, by a) extending the applicability of the theoretical solutions to nonoptimum or non-ideal conditions and b) providing the designer with more insight information on the tradeoffs involved with his design, since this analysis can point out the combined effects of all the parameters which strongly affect the amplifier performance. Furthermore, since this is a generalized case, very useful information for other circuit configurations can be obtained as well. This is done in [5] where the effect of the parasitic device shunting capacitance on the performance of the circuit configurations of [6] and [7] is investigated.

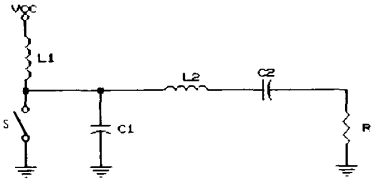


Fig. 1. A generalized Class E amplifier configuration.

The analysis is straightforward, but time consuming, and the resulting equations are given in Appendix A. In addition, the general analysis procedure is discussed and the analysis and design results for the case of the finite dc-feed inductance, shunt capacitance, series-tuned Class E circuit configuration operating at a 50-percent duty-ratio are discussed and are given in the form of families of curves.

Finally, four extreme design cases have been selected and were constructed in the laboratory, in order to verify the theoretical predictions. Experimental results show an excellent agreement with theory, and the observed differences were typically under 4 percent.

II. Analysis and Design Procedure

The assumptions made in this analysis are the following.

a) The switching device has zero switching times.[1]

b) The parasitic elements are considered to be constant and are included in the load-network components (e.g., the active parasitic capacitance is absorbed by the shunting capacitance C_1, etc.).

c) The load is fixed and purely ohmic. The fixed-load assumption is required for Class E amplifier optimum operation. However, the purely ohmic load assumption is

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[1]Although it would be very interesting to include nonzero switching times, this would make the theoretical analysis too complex to be carried out. However, a good approximation may be obtained by combining the results given here with those given in [8] and [9].

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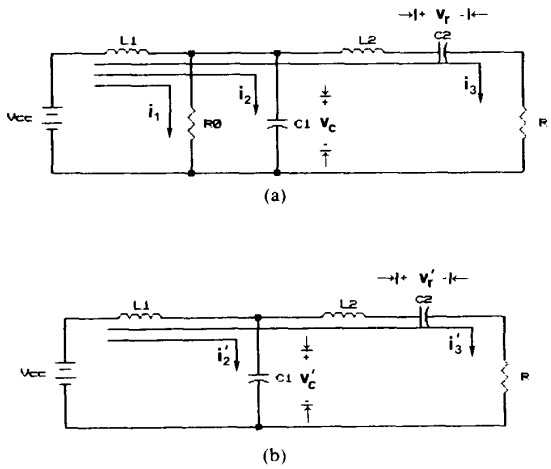


Fig. 2. Equivalent circuit: (a) when the switch is ON; (b) when it is OFF.

made here only for convenience, since it is possible to separate any load impedance into a purely ohmic portion and a purely reactive portion which may be absorbed by the load-network components such that the theoretically predicted load angle, for the purely ohmic-load case is preserved.

In accordance with the above assumptions, two equivalent circuits of the generalized Class E amplifier configuration for the ON and OFF states of the switch were formed, as shown in Fig. 2(a) and (b), respectively. The differential equations which describe these two circuits are as follows.

For the ON state,

$$\left.\begin{aligned} \frac{di_1(t)}{dt}+\frac{di_2(t)}{dt}+\frac{di_3(t)}{dt}&=\frac{i_1(t)R_0-V_{cc}}{L_1}\\ \frac{1}{C_1}\int_{-\infty}^{t} i_2(\tau)\,d\tau&=i_1(t)R_0\\ L_2\frac{di_3(t)}{dt}&=i_1(t)R_0-v_r(t)-i_3(t)R\\ v_r(t)&=\frac{1}{C_2}\int_{-\infty}^{t} i_3(\tau)\,d\tau \end{aligned}\right\} \quad (1)$$

where R_0 is the active device ON resistance and for the OFF[2] state:

$$\left.\begin{aligned} \frac{di_2't)}{dt}+\frac{di_3'(t)}{dt}&=\frac{V_{cc}-V_c'(t)}{L_1}\\ L_2\frac{di_3'(t)}{dt}&=v_c'(t)-v_r'(t)-i_3'(t)R\\ v_r'(t)&=\frac{1}{C_2}\int_{-\infty}^{t} i_3'(\tau)\,d\tau\\ v_c'(t)&=\frac{1}{C_1}\int_{-\infty}^{t} i_2'(\tau)\,d\tau. \end{aligned}\right\} \quad (2)$$

[2]Circuit parameters are denoted primed for the OFF state of the switch.

Transforming (1) and (2) to the s-domain and solving for the state variables, we get

$$\mathbb{X}(s)=\frac{\mathbb{N}(s)}{\boldsymbol{D}(s)} \quad (3)$$

and

$$\mathbb{X}'(s)=\frac{\mathbb{N}'(s)}{\boldsymbol{D}'(s)} \quad (4)$$

where

$\mathbb{X}^T(s)$ $[I_1(s)\ \ I_2(s)\ \ I_3(s)\ \ V_c(s)\ \ V_r(s)]$

$\mathbb{X}'^T(s)$ $[I_1'(s)\ \ I_2'(s)\ \ I_3'(s)\ \ V_c'(s)\ \ V_r'(s)]$

$\mathbb{N}(s)$ and $\mathbb{N}'(s)$ vectors whose elements are functions of s, of the circuit element values and of the circuit initial conditions,

$\boldsymbol{D}(s)$ and $\boldsymbol{D}'(s)$ fourth-order polynomials of s with coefficients which are functions of the circuit element values.

Since it is very difficult to express the roots of $\boldsymbol{D}(s)$ and $\boldsymbol{D}'(s)$ analytically, in order to carry out the inverse Laplace transform using residues, it is assumed that the roots are known and one of the following combinations is true: a) four real roots, b) two real roots and a pair of complex conjugates or c) two pairs of complex-conjugate roots. After transforming back to the t-domain, the state variables are expressed as functions of these roots, which in turn are functions of the circuit element values. Given these values, the roots can be calculated numerically.

The solutions of (1) and (2) are expressed in the form:

$$\mathbb{X}(t)=\mathbf{A}(t)\mathbb{X}(0)+\mathbb{B}(t)V_{cc} \quad (5)$$

$$\mathbb{X}'(t)=\mathbf{A}'(t)\mathbb{X}'(0)+\mathbb{B}'(t)V_{cc} \quad (6)$$

where $\mathbf{A}(t)$, $\mathbf{A}'(t)$, $\mathbb{B}(t)$, and $\mathbb{B}'(t)$ are 5×5, 4×4, 5×1, and 4×1 matrices, respectively, whose elements are functions of time and of the circuit element values.

It is evident that there are three different sets of equations in the form of (5) and (6) for each of the three possible root cases mentioned previously.

The next step is to eliminate $\mathbb{X}(0)$ and $\mathbb{X}'(0)$ from (5) and (6). This is done using the following boundary conditions which hold when the switch changes state:

$$\left.\begin{aligned} i_1(0)+i_2(0)&=i_2'(t_1)\\ i_3(0)&=i_3'(t_1)\\ v_c(0)&=v_c'(t_1)\\ v_r(0)&=v_r'(t_1) \end{aligned}\right\} \quad (7)$$

and

$$\left.\begin{aligned} i_2'(0)&=i_1(t_2)+i_2(t_2)\\ i_3'(0)&=i_3(t_2)\\ v_c'(0)&=v_c(t_2)\\ v_r'(0)&=v_r(t_2) \end{aligned}\right\} \quad (8)$$

where t_1 and t_2 are the instants when the switch turns OFF and ON, respectively.

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The combination of (5)–(8) leads to a linear system of equations in the form:

$$\mathbb{W}(t)\mathbb{X}(0) = \mathbb{Z}(t). \tag{9}$$

The solution of (9) is carried out numerically and it provides the circuit steady-state initial conditions as functions of the circuit element values. With these values known, the operation of the circuit can be fully analyzed.

The next step is to design the circuit, i.e., to find the proper circuit element values so that a desired set of design conditions is satisfied. One such set of conditions is [1]

$$v_c'(t_2) = 0 \quad \text{and} \quad \left.\frac{dv_c'(t)}{dt}\right|_{t=t_2} = 0 \tag{10}$$

where t_2 the instant when the switch turns ON. Design conditions (10) form a nonlinear system of equations. Direct-search optimization techniques, which were used to find the solution, are preferred because the calculation of the partial derivatives required by other methods is very complicated to be carried out analytically, and is time consuming if it is done numerically. The simplex minimization [11] method was selected out of direct optimization techniques mainly because of its robustness, despite the fact that it is rather slow, compared to other methods (e.g., Hooke–Jeeves). With two design conditions considered, only two circuit element values can be calculated, while all the others are handled as parameters.

III. Presentation and Discussion of Results

The nonlinear system of equations (10) is solved for a 50-percent duty ratio, i.e., $t_2 = \pi$, to provide the design values for a) the normalized detuning portion of L_2, defined by

$$L_2' \equiv \frac{\omega L_2}{R} - Q_L \tag{11}$$

where[3]

$$Q_L \equiv (1/\omega)C_2R$$

and b) the shunting capacitance C_1, with Q_L, R_0, and L_1 are defined as parameters. The results are shown for the normalized case $R = 1$, $V_{cc} = 1$, and $\omega = 1$.

Fig. 3(a) and (b) show the normalized design values as functions of R_0 normalized to R, for four different Q_L values (0.1, 1, 5, and 20) and for three different L_1 design cases (100, 10, and 1). When $L_1 = 100$, the case corresponds to using an RF-choke. Fig. 3(c) and (d), on the other hand, show the design values as functions of L_1, for $R_0 = 0.01$.

It can be observed that the design values are slightly affected by R_0 variations for small R_0 values, while for values larger than 0.1, the dependence is much stronger. Design values are strongly affected by L_1, especially for low L_1 values. It is also observed that the dependence of C_1 on Q_L becomes stronger as L_1 decreases, while the opposite holds true for L_2'.

It should be stated that the design conditions cannot be met for very low L_1 values, typically under 0.6, slightly decreasing with Q_L. The limit for the case of $Q_L = 20$ at 50-percent duty cycle, was found to be about 0.6 which agrees with the value given by Zulinski in [2] obtained for the case of infinite Q_L at 50-percent duty ratio.

Fig. 4 shows how the calculated device stresses (i.e., $v_c(\max)$ and $i_1(\max)$) vary with R_0 and L_1 with Q_L as a parameter. It is seen that the device stress decreases as R_0 increases and that this decrease is faster for low L_1 values. Furthermore, the effect of Q_L on the device stress is more significant at low R_0 values. On the other hand, low Q_L values cause larger $v_c(\max)$, while the opposite is true for $i_1(\max)$, except for very low values of L_1. It is also noted in Fig. 4(c) that $v_c(\max)$ is an insensitive function of L_1 for given R_0 and Q_L. Finally, it is seen that $i_1(\max)$ increases rapidly as L_1 decreases and this increase is faster for low Q_L values. This is expected, since L_2' decreases towards zero with L_1, as shown in Fig. 3(c), which means that the amplifier's optimum operating point moves towards the peak of the resonance curve, thus presenting less impedance to the voltage source.

Fig. 5 depicts the variation of input and output power (P_{in} and P_{out}) with R_0 and L_1, taking Q_L as a parameter. P_{in} and P_{out} were calculated numerically from the equations:

$$P_{in} = \frac{V_{cc}}{2\pi}\left[\int_0^{\pi}[i_1(t) + i_2(t) + i_3(t)]\,dt + \int_0^{\pi}[i_2'(t) + i_3'(t)]\,dt\right] \tag{12}$$

and

$$P_{out} = \frac{R}{2\pi}\left[\int_0^{\pi} i_3^2(t)\,dt + \int_0^{\pi} i_3'^2(t)\,dt\right]. \tag{13}$$

Both P_{in} and P_{out} decrease as R_0 increases. This is expected since larger R_0 values cause a larger voltage drop at the active device. This voltage drop is subtracted from V_{cc} to provide the effective input voltage, as explained in [8]. Furthermore, since this voltage drop is larger at low L_1 values (where $i_1(\max)$ is larger) the decrease of both P_{in} and P_{out} is expected to be larger. It is also observed that Q_L affects less the amplifier performance (and consequently, P_{in} and P_{out} as L_1 decreases and R_0 increases. Finally, P_{in} and P_{out} increase rapidly as L_1 decreases (Fig. 5(c) and (d)). This means that designs with low L_1 should be selected in cases where high power is needed, but the design specifications require a small V_{cc} and a large R.

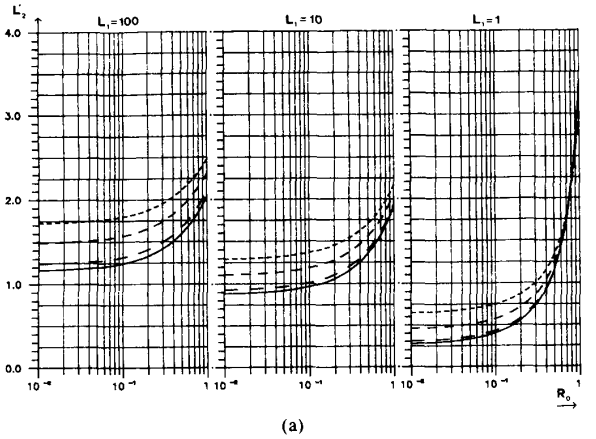
Fig. 6 shows the efficiency, η, and the power output capability (P_{max}) as functions of R_0 and L_1 with Q_L as a parameter, where

$$P_{max} = \frac{P_{out}}{v_c(\max)i_1(\max)} \tag{14}$$

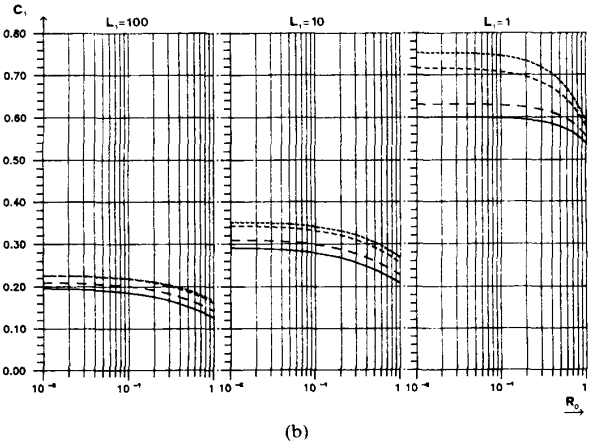
as expected, efficiency decreases as R_0 increases. The

[3] Note that this definition of Q_L is the same as in [3] and [4], but [1] defines Q_L as $\omega L_2/R$.

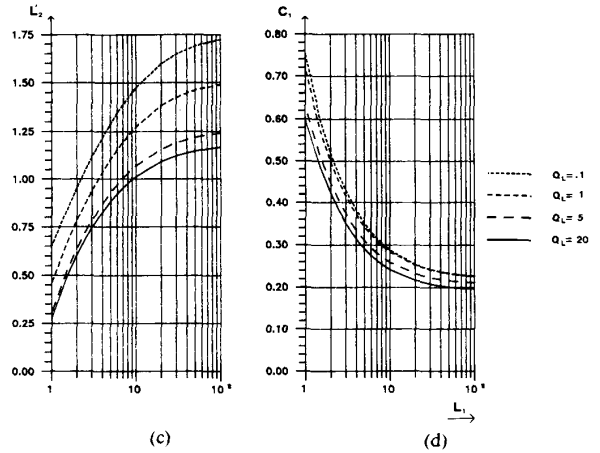
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(a)

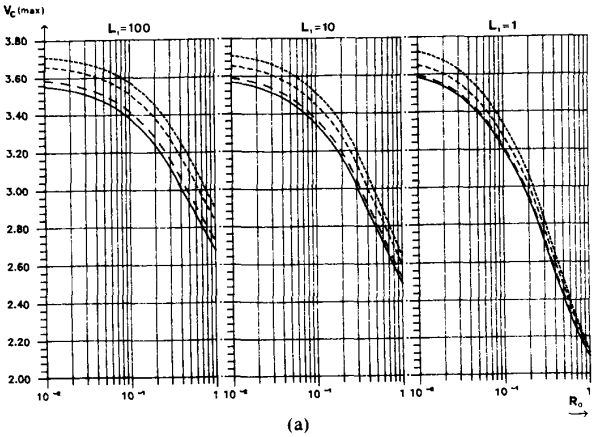


(b)

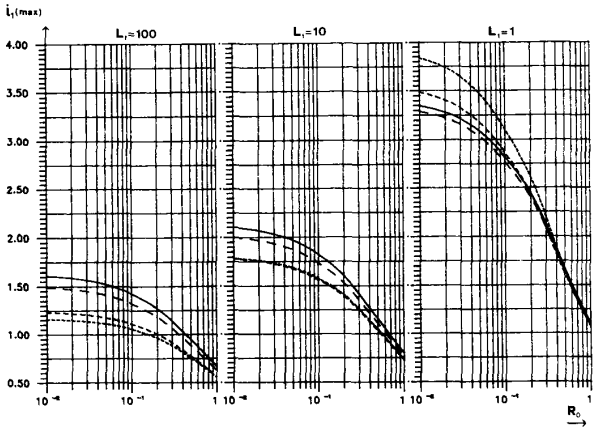


(c) (d)

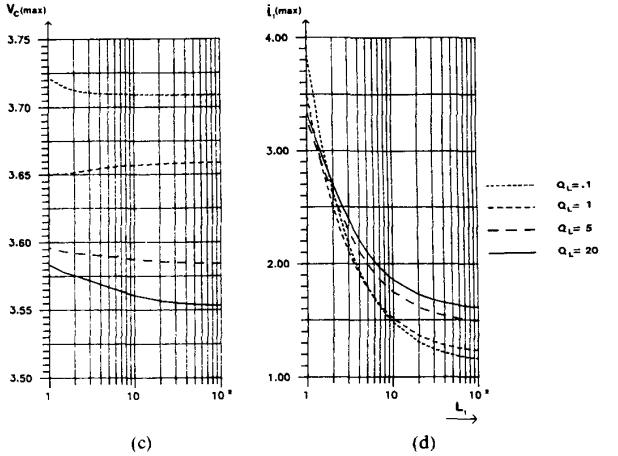
Fig. 3. Calculated normalized design values of (a) L_2' and (b) C_1 versus R_0 for $L_1 = 100$, 10, and 1, (c) L_2' and (d) C_1 versus L_1 at $R_0 = 0.01$, all taken for $Q_L = 0.1$, 1, 5, and 20.



(a)



(b)



(c) (d)

Fig. 4. Normalized device stress of (a) v_c(max) and (b) i_1(max) versus R_0 for $L_1 = 100$, 10, and 1, (c) v_c(max) and (d) i_1(max) versus L_1 at $R_0 = 0.01$, all taken for $Q_L = 0.1$, 1, 5, and 20.

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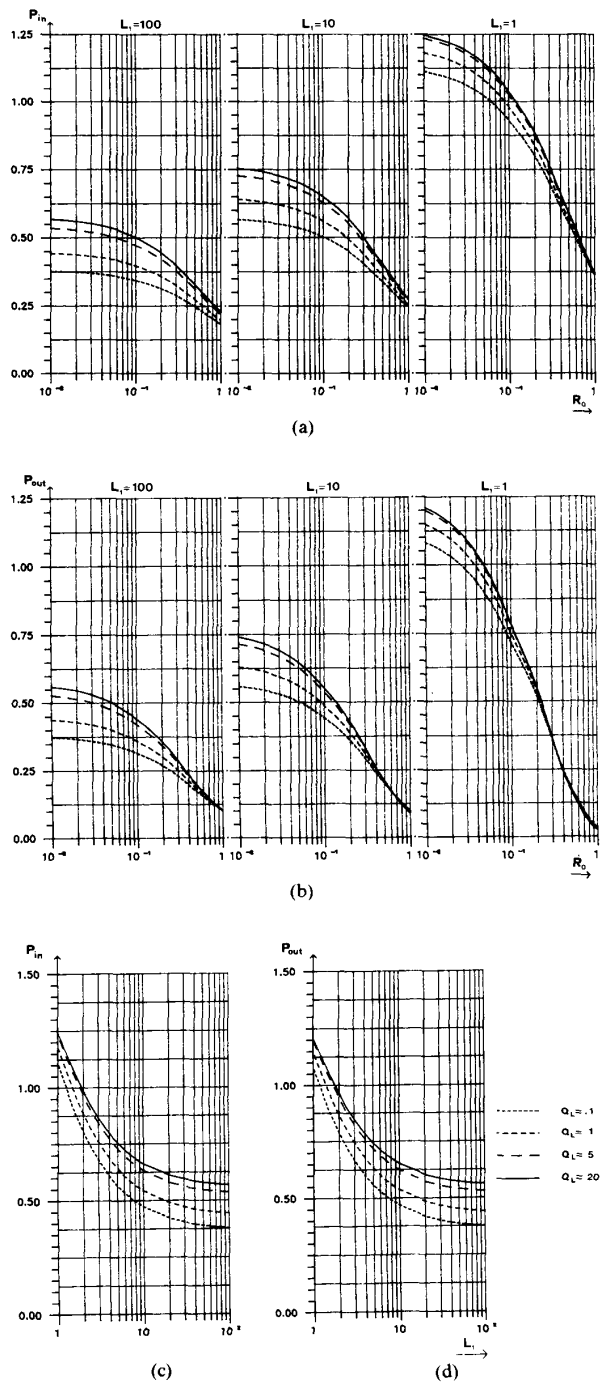


Fig. 5. Normalized power input and output. (a) P_{in} and (b) P_{out} versus R_0 for $L_1 = 100$, 10, and 1, (c) P_{in} and (d) P_{out} versus L_1 at $R_0 = 0.01$, all taken for $Q_L = 0.1$, 1, 5, and 20.

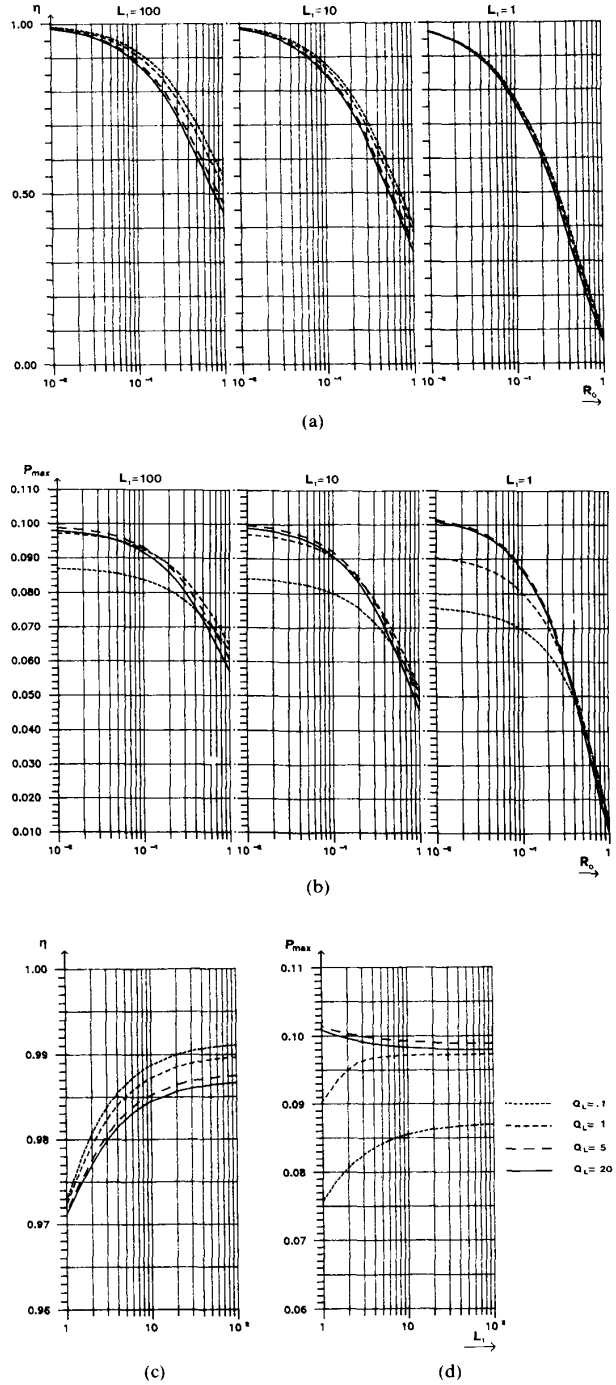


Fig. 6. Efficiency and power output capability; (a) η and (b) P_{max} versus R_0 for $L_1 = 100$, 10, and 1, (c) η and (d) P_{max} versus L_1 at $R_0 = 0.01$, all taken for $Q_L = 0.1$, 1, 5, and 20.

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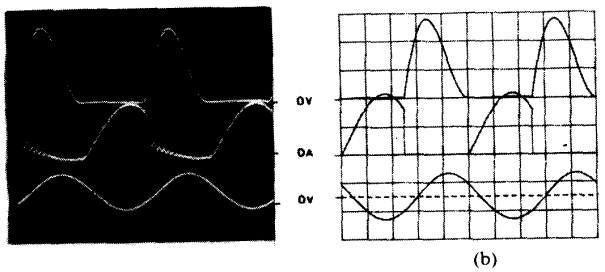


Fig. 7. (a) Experimental and (b) calculated waveforms for $L_1 = 100$ and $Q_L = 20$ with $R_0 = 0.04$. Upper trace: v_c 20 V/div. Middle trace: i_1 0.5 A/div. Lower trace: $v_0 = i_3 R$ 20 V/div. Time scale 0.2 μs/div.

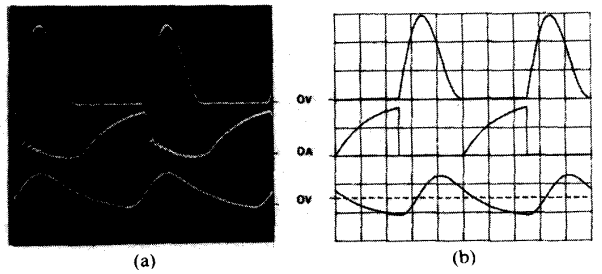


Fig. 8. (a) Experimental and (b) calculated waveforms for $L_1 = 100$ and $Q_L = 0.1$ with $R_0 = 0.04$. Scales as in Fig. 7.

power dissipated at R_0 is larger when L_1 is small, because of the corresponding larger i_1(max) value. It is also seen from Fig. 6(c) that there is an approximately 1.5-percent decrease in efficiency, for the full L_1 scale at $R_0 = 0.01$. However, this decrease is much larger for larger R_0 values, which means that R_0 should certainly be taken into account in designs with low L_1 values. It is seen that the amplifier efficiency is less sensitive to Q_L at low L_1 values, while the efficiency is higher at low values of Q_L. As expected, P_{max} also decreases as R_0 increases, and this decrease is faster with low L_1 values. In Fig. 6(d), it is observed that P_{max} increases slightly as L_1 decreases for high Q_L values, but it decreases significantly for low Q_L and L_1 values.

IV. Experimental Verification

Four different Class E amplifier circuits were constructed to verify the results given above, with Q_L and normalized L_1 values of a) $Q_L = 20$, $L_1 = 100$; b) $Q_L = 0.1$, $L_1 = 100$; c) $Q_L = 20$, $L_1 = 1$; and d) $Q_L = 0.1$, $L_1 = 1$. The switching device used was a VMOS VN64GA driven by a 50-percent duty cycle, 0–10-V rectangular pulse at 1 MHz. RF power measurements were made using a Fluke Model 8921A true rms voltmeter with a 0.7-percent accuracy in the frequency and voltage ranges used. The circuit component values were selected to be close to those predicted theoretically.

Fig. 7(a) shows the switch voltage and current[4] as well as the output voltage, for $Q_L = 20$ and $L_1 = 100$. It is clearly seen that the design conditions are met. Fig. 7(b) shows the theoretically predicted waveforms,[5] while Table I summarizes the values of the theoretical and experimental results. A comparison shows differences under 9 percent. The biggest differences are observed in v_c(max) and i_1(max) which are very sensitive to small changes in the design values. As expected, efficiency is slightly smaller than the theoretically predicted, since there also exist other reasons which cause power dissipation, such as nonzero switching times, lead inductances and resistances which were not accounted for.

[4] The approximately 20-ns delay in the current waveform is caused by the Tektronix model AM503 current probe used.

[5] The current difference between Fig. 7(a) and (b) and all the subsequent figures, when the switch is OFF, is due to the current flowing through the active device parallel output parasitic capacitance.

TABLE I
$Q_L = 20$; $L_1 = 100$

Circuit Element	Normaliz.[1] Values	Calcul.[2] Values	Measured Values	Parasitic Elements	Total[3] Value	Diff.[4] %
			CIRCUIT ELEMENT VALUES			
Vcc (V)	1.00	16.00	16.00		16.00	0.00
R (Ω)	1.00	22.50	20.50	2.15	22.65	0.67
R0 (Ω)	0.04	0.80	0.80		0.80	0.00
L2' (μH)	1.19071	--	--		--	--
L2 (μH)	21.19071	75.88	75.80	2.15 Ω	75.80	0.11
L1 (μH)	100.00000	358.10	360.00	0.3 Ω	360.00	0.53
C1 (pF)	0.19151	1354.65	966.00	400.00	1366.00	0.84
C2 (pF)	0.05000	353.68	350.00		350.00	1.04
			CIRCUIT PERFORMANCE VALUES			
Vc(max) (V)	3.48966	55.834	59.00		59.000	5.67
I1(max) (A)	1.54179	1.096	1.00		1.000	8.79
P(diss.@ L1)(W)				0.522		--
P(diss.@ L2)(W)				0.111		--
Pout (W)	0.51500	5.860	4.976	0.633	5.609	4.28
Pin (W)	0.54286	6.177	6.080		6.080	1.56
Efficiency	0.94000	0.94000	0.818		0.9225	1.86
Pmax	0.09600	0.09600	0.0843		0.0951	0.97

[1] Values Obtained from Fig. 3

[2] Calculated From the normalized values

[3] Measured plus parasitic values.

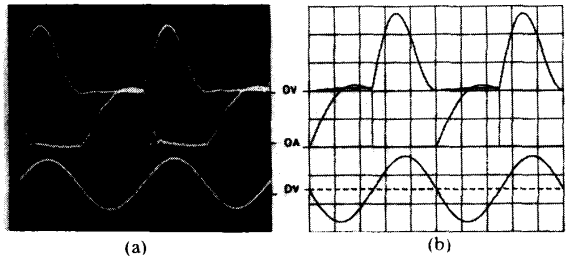
[4] Percentage difference calculated as |Calculated-Total|/Calculated *100%

TABLE II
$Q_L = 0.1$; $L_1 = 100$

Circuit Element	Normalized Values	Calcul. Values	Measured Values	Parasitic Elements	Total Value	Diff. %
			CIRCUIT ELEMENT VALUES			
Vcc (V)	1.00	16.00	16.00		16.00	0.00
R (Ω)	1.00	20.80	20.50	0.30	20.80	0.00
R0 (Ω)	0.04	0.80	0.80		0.80	0.00
L2' (μH)	1.74636	--	--		--	--
L2 (μH)	1.84636	6.11	6.23	0.3 Ω	6.23	1.93
L1 (μH)	100.00000	331.04	330.00	0.3 Ω	330.00	0.31
C1 (pF)	0.22267	1703.82	1303.00	400.00	1703.00	0.05
C2 (pF)	10.00000	76516.80	76500.00		76500.00	0.02
			CIRCUIT PERFORMANCE VALUES			
Vc(max) (V)	3.65954	58.553	60.500		60.500	3.33
I1(max) (A)	1.12012	0.862	0.830		0.830	3.67
P(diss.@ L1)(W)				0.057		
P(diss.@ L2)(W)				0.081		
Pout (W)	0.35170	4.329	3.910	0.138	4.048	6.49
Pin (W)	0.36437	4.485	4.300		4.300	4.12
Efficiency	0.96523	0.96523	0.9093		0.9414	2.47
Pmax	0.08580	0.08580	0.0779		0.0806	6.05

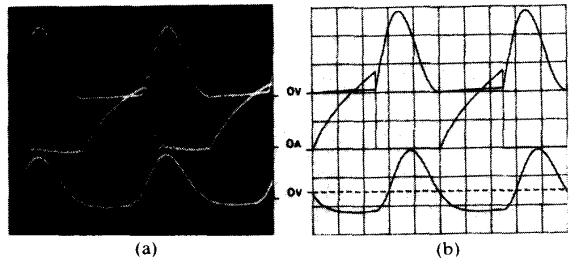
Fig. 8(a) shows the amplifier waveforms for the case that $Q_L = 0.1$ and $L_1 = 100$. It s noted that the design conditions are met and that the current waveform tends to become a ramp, while the output voltage contains many harmonics. Fig. 8(b) depicts the same waveforms, calculated theoretically. A summary of the theoretical and experimental values is presented in Table II. All differences here are under 7 percent. The efficiency is higher than that

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(a) (b)

Fig. 9. (a) Experimental and (b) calculated waveforms for $L_1 = 1$ and $Q_L = 20$ with $R_0 = 0.04$. Upper trace: v_c 20 V/div. Middle trace: i_1 1 A/div. Lower trace: $v_0 = i_3 R$ 20 V/div. Time scale 0.2 μs/div.



(a) (b)

Fig. 10. (a) Experimental and (b) calculated waveforms for $L_1 = 1$ and $Q_L = 0.1$ with $R_0 = 0.04$. Scales as in Fig. 9.

TABLE III
$Q_L = 20$; $L_1 = 1$

Circuit Element	Normalized Values	Calcul. Values	Measured Values	Parasitic Elements	Total Valuee	Diff. %
			CIRCUIT ELEMENT VALUES			
Vcc (V)	1.00	16.00	16.00		16.00	0.00
R (Ω)	1.00	22.85	20.50	2.35	22.85	0.00
R0 (Ω)	0.04	0.80	0.80		0.80	0.00
L2 (μH)	0.31600	--	--		--	--
L2 (μH)	20.31600	73.88	73.00	2.35 Ω	73.00	1.20
L1 (μH)	1.00000	3.64	3.58	0.27 Ω	3.58	1.56
C1 (pF)	0.59932	4174.40	3774.00	400.00	4174.00	0.01
C2 (pF)	0.05000	348.26	348.00		348.00	0.07
			CIRCUIT PERFORMANCE VALUES			
Vc(max) (V)	3.43997	55.039	55.500		55.500	0.84
I1(max) (A)	3.14905	2.205	2.200		2.200	0.23
P(diss.@ L1)(W)				1.158		
P(diss.@ L2)(W)				0.219		
Pout (W)	1.04050	11.657	10.100	1.477	11.477	1.54
Pin (W)	1.16722	13.077	13.000		13.000	0.59
Efficiency	0.89144	0.89144	0.7769		0.8829	0.96
Pmax	0.09606	0.09606	0.0827		0.0940	2.15

TABLE IV
$Q_L = 0.1$; $L_1 = 1$

Circuit Element	Normalized Values	Calcul. Values	Measured Values	Parasitic Elements	Total Value	Diff. %
			CIRCUIT ELEMENT VALUES			
Vcc (V)	1.00	16.00	16.00		16.00	0.00
R (Ω)	1.00	20.68	20.50	0.18	20.68	0.01
R0 (Ω)	0.04	0.80	0.80		0.80	0.00
L2 (μH)	0.67967	--	--		--	--
L2 (μH)	0.77967	2.57	2.57	0.18 Ω	2.57	0.03
L1 (μH)	1.00000	3.29	3.27	0.26 Ω	3.27	0.65
C1 (pF)	0.75044	5775.43	5375.00	400.00	5775.00	0.01
C2 (pF)	10.00000	76960.80	77000.00		77000.00	0.05
			CIRCUIT PERFORMANCE VALUES			
Vc(max) (V)	3.57484	57.197	58.000		58.000	1.40
I1(max) (A)	3.55405	2.750	2.600		2.600	5.45
P(diss.@ L1)(W)				0.208		
P(diss.@ L2)(W)				0.096		
Pout (W)	0.93540	11.579	10.830	0.304	11.134	3.85
Pin (W)	1.04179	12.896	12.800		12.800	0.75
Efficiency	0.89789	0.89789	0.8461		0.8699	3.12
Pmax	0.07362	0.07362	0.0718		0.0738	0.29

obtained when $Q_L = 20$, since at this low value of Q_L, the power dissipated at the parasitic resistances of L_1 and L_2 is much less, thus increasing significantly the overall amplifier efficiency.

Fig. 9 shows the experimental and the theoretically predicted waveforms and Table III presents the measured and theoretical values for the case that $Q_L = 20$ and $L_1 = 1$. The agreement here is excellent with differences smaller than 2.2 percent for every value shown. It should be noted that the power output is now more than double the power output obtained at the high-L_1, high-Q_1 case discussed above. However, the efficiency is about 3 percent less.

Finally, Fig. 10 and Table IV deal with the last case where $Q_L = 0.1$ and $L_1 = 1$. Again the agreement between experimental and theoretical values is very good with differences less than 6 percent. As in the previous case with $Q_L = 0.1$, the current waveform becomes a ramp and the output voltage contains many harmonics.

V. Conclusions

A generalized Class E amplifier configuration has been analyzed. The analysis method used takes into account all the circuit components which strongly affect the amplifier performance, i.e., the dc-feed inductance, the Q_L factor of the series-tuned circuit, and the switching-device ON resistance. The analysis described may yield results for most of the other Class E amplifier configurations studied so far, by simply changing the design conditions and reducing to zero the values of the missing components. The results given for the finite dc-feed inductance, series-tuned, shunt-capacitor amplifier configuration case provide a clear insight of the overall amplifier performance and the way it is affected by R_0, L_1 and Q_L. It is seen that R_0 should be taken into account, especially when it is larger than $0.1R$. Moreover, the power output increases substantially when L_1 is small and this happens because of the smaller detuning needed, for the series-tuned circuit. This makes designs with small L_1 more attractive in RF transmitter applications, where the amplitudes of the two sidebands should not be substantially different. Designs with small L_1 are also attractive in cases where both the input voltage and the load resistance are specified and higher power output is needed. It is also seen that Q_L affects the amplifier performance less when L_1 is small and R_0 is large. Finally, it is observed that P_{max} increases as L_1 decreases, but this is true only for large values of Q_L, while the opposite is true for small Q_L values.

Laboratory results show a very close agreement with the theoretically predicted values for the four implemented experimental Class E amplifier circuits which were designed using extreme values for Q_L and L_1.

Appendix A
The Resulting Equations

The resulting equations given in this appendix concern the three most common root cases of $\boldsymbol{D}$ and $\boldsymbol{D}'$ of (3) and (4). The exclusion of the other three root cases is not

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expected to cause substantial problems since these cases are rarely used and only during the computation of the component values of extreme design cases.

i) *When The Switch is ON*

The denominator polynomial of (3) is expressed as

$$D(s) = s^4 + As^3 + Bs^2 + \Gamma s + \Delta \qquad \text{(A.1)}$$

where

$$A = R/L_2 + 1/(C_1R_0),$$
$$B = R/(L_2C_1R_0) + 1/(L_1C_1) + 1/(L_2C_2) + 1/(L_2C_1)$$
$$\Gamma = 1/(L_2C_1C_2R_0) + R/(L_1L_2C_1),$$

and

$$\Delta = 1/(L_1L_2C_1C_2). \qquad \text{(A.2)}$$

Equation (5) is expressed by

$$\left.\begin{aligned} i_1(t) &= F_2(i_1(0)+i_2(0)) + F_3i_3(0) + F_4v_c(0) + F_5v_r(0) + F_6V_{cc} \\ i_2(t) &= G_2(i_1(0)+i_2(0)) + G_3i_3(0) + G_4v_c(0) + G_5v_r(0) + G_6V_{cc} \\ i_3(t) &= X_2(i_1(0)+i_2(0)) + X_3i_3(0) + X_4v_c(0) + X_5v_r(0) + X_6V_{cc} \\ v_c(t) &= U_2(i_1(0)+i_2(0)) + U_3i_3(0) + U_4v_c(0) + U_5v_r(0) + U_6V_{cc} \\ v_r(t) &= W_2(i_1(0)+i_2(0)) + W_3i_3(0) + W_4v_c(0) + W_5v_r(0) + W_6V_{cc} \end{aligned}\right\} \qquad \text{(A.3)}$$

where

$$F_2 = \mathcal{f}_2/(C_1C_2L_2R_0), \quad F_3 = \mathcal{f}_3/(C_1C_2L_2R_0), \quad F_4 = \mathcal{f}_4/(C_2L_2R_0)$$
$$F_5 = \mathcal{f}_5/(C_1L_2R_0), \quad F_6 = (\mathcal{f}_6 + k_1)/(C_1C_2L_1L_2R_0)$$
$$G_2 = \mathcal{g}_2/(C_2L_2), \quad G_3 = \mathcal{g}_3/(C_2L_2), \quad G_4 = \mathcal{g}_4/(L_1L_2C_2R_0)$$
$$G_5 = \mathcal{g}_5/L_2, \quad G_6 = \mathcal{g}_6/(C_2L_1L_2)$$
$$X_2 = \mathcal{x}_2/(C_1L_2), \quad X_3 = \mathcal{x}_3/(C_1L_1L_2R_0), \quad X_4 = G_5$$
$$X_5 = \mathcal{x}_5/(C_1L_1L_2R_0), \quad X_6 = \mathcal{x}_6/(C_1L_1L_2)$$
$$W_2 = \omega_2/(C_1C_2L_2), \quad W_3 = \omega_3/(C_1C_2L_1L_2R_0), \quad W_4 = \omega_4/(L_2C_2)$$
$$W_5 = 1 + \omega_5/(C_1C_2L_1L_2R_0), \quad W_6 = \omega_6/(C_1C_2L_1L_2)$$
$$U_2 = \mathcal{u}_2/(C_1C_2L_2), \quad U_3 = \mathcal{u}_3/(C_1C_2L_2), \quad U_4 = 1 + \mathcal{u}_4/(C_1C_2L_1L_2R_0)$$
$$U_5 = W_4C_2/C_1, \quad U_6 = \mathcal{u}_6/(C_1C_2L_1L_2)$$
$$\mathcal{f}_j = f_{j1}s_a + f_{j2}s_b + f_{j3}s_c + f_{j4}s_d, \quad \mathcal{g}_j = g_{j1}s_a + g_{j2}s_b + g_{j3}s_c + g_{j4}s_d$$
$$\mathcal{x}_j = x_{j1}s_a + x_{j2}s_b + x_{j3}s_c + x_{j4}s_d, \quad \omega_j = w_{j1}r_a + w_{j2}r_b + w_{j3}r_c + w_{j4}r_d$$
$$\mathcal{u}_j = u_{j1}r_a + u_{j2}r_b + u_{j3}r_c + u_{j4}r_d \quad \text{and} \quad j = 2 \cdots 6.$$

The coefficients $s_{a\cdots d}$, $r_{a\cdots d}$, f_{ji}, g_{ji}, x_{ji}, w_{ji} and u_{ji} are functions of the circuit element values and of the roots of the polynomial given in (A.1):

a) When all four roots are real $(\rho_1, \rho_2, \rho_3, \rho_4)$:

$$s_a = e^{\rho_1 t}/D_1, \quad s_b = e^{\rho_2 t}/D_2, \quad s_c = e^{\rho_3 t}/D_3, \quad s_d = e^{\rho_4 t}/D_4$$
$$r_a = (e^{\rho_1 t} - 1)/D_1, \quad r_b = (e^{\rho_2 t} - 1)/D_2, \quad r_c = (e^{\rho_3 t} - 1)/D_3, \quad r_d = (e^{\rho_4 t} - 1)/D_4$$

where

$$D_1 = [(\rho_1 - \rho_2)(\rho_1 - \rho_3)(\rho_1 - \rho_4)], \quad D_2 = [(\rho_2 - \rho_1)(\rho_2 - \rho_3)(\rho_2 - \rho_4)]$$
$$D_3 = [(\rho_3 - \rho_1)(\rho_3 - \rho_2)(\rho_3 - \rho_4)], \quad D_4 = [(\rho_4 - \rho_1)(\rho_4 - \rho_2)(\rho_4 - \rho_3)]$$

and

$$f_{2i} = C_2L_2\rho_i^2 + RC_2\rho_i + 1, \quad f_{3i} = RC_2\rho_i + 1, \quad f_{4i} = \rho_i f_{2i}, \quad f_{5i} = \rho_i$$
$$f_{6i} = f_{2i}/\rho_i, \quad g_{2i} = f_{4i}, \quad g_{3i} = \rho_i f_{3i}$$
$$g_{4i} = -\rho_i^3 L_1L_2C_2 - (RL_1C_2 + L_2C_2R_0 + L_1C_2R_0)\rho_i^2 - (L_1 + RC_2R_0)\rho_i - R_0, \; g_{5i} = \rho_i^2$$
$$g_{6i} = f_{2i}, \quad x_{2i} = f_{5i}, \quad x_{3i} = \rho_i(C_1L_1L_2R_0\rho_i^2 + L_1L_2\rho_i + R_0(L_1 + L_2))$$
$$x_{5i} = -\rho_i^2 C_1L_1R_0 - \rho_i L_1 - R_0, \quad x_{6i} = w_{2i} = 1, \quad w_{3i} = x_{3i}/\rho_i, \quad w_{4i} = \rho_i$$
$$w_{5i} = x_{5i}/\rho_i, \quad w_{6i} = 1/\rho_i, \quad u_{2i} = f_{2i}, \quad u_{3i} = f_{3i}, \quad u_{41} = g_{4i}/\rho_i, \quad u_{6i} = f_{6i}$$

where $i = 1 \cdots 4$ and $k_1 = 1/(\rho_1\rho_2\rho_3\rho_4)$.

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b) When the roots are both real and complex $(\rho_1, \rho_2, \alpha \pm j\beta)$:

$$s_a = e^{\rho_1 t}/\rho_1(\rho_1-\rho_2)\big((\rho_1-\alpha)^2+\beta^2\big), \quad s_b = e^{\rho_2 t}/\rho_2(\rho_2-\rho_1)\big((\rho_2-\alpha)^2+\beta^2\big)$$
$$s_c = e^{\alpha t}\cos(\beta t)/\big(s_{15}^2+s_{10}^2\big), \quad s_d = e^{\alpha t}\sin(\beta t)/\big(s_{15}^2+s_{10}^2\big)$$
$$\sigma_a = s_a\rho_1, \quad \sigma_b = s_b\rho_2, \quad \sigma_c = e^{\alpha t}\cos(\beta t)/\big(s_{24}^2+s_{28}^2\big)$$
$$\sigma_d = e^{\alpha t}\sin(\beta t)/\big(s_{24}^2+s_{28}^2\big), \quad r_a = (e^{\rho_1 t}-1)/\rho_1(\rho_1-\rho_2)\big((\rho_1-\alpha)^2+\beta^2\big)$$
$$r_b = (e^{\rho_2 t}-1)/\rho_2(\rho_2-\rho_1)\big((\rho_2-\alpha)^2+\beta^2\big)$$
$$r_c = \big(e^{\alpha t}(\alpha\cos(\beta t)+\beta\sin(\beta t))-\alpha\big)/\big(s_{24}^2+s_{28}^2\big)(\alpha^2+\beta^2)$$
$$r_d = \big(e^{\alpha t}(\alpha\sin(\beta t)-\beta\cos(\beta t))+\beta\big)/\big(s_{24}^2+s_{28}^2\big)(\alpha^2+\beta^2)$$
$$f_{21} = \rho_1\big(\rho_1^2L_2C_2+\rho_1RC_2+1\big), \quad f_{22} = \rho_2\big(\rho_2^2L_2C_2+\rho_2RC_2+1\big)$$
$$f_{23} = L_2C_2s_{12}+RC_2s_{13}+s_{14}, \quad f_{24} = -L_2C_2s_{17}-RC_2s_{18}-s_{19}$$
$$f_{31} = \rho_1(RC_2\rho_1+1), \quad f_{32} = \rho_2(RC_2\rho_2+1), \quad f_{33} = RC_2s_{13}+s_{14}, \quad f_{34} = -RC_2s_{18}-s_{19}$$
$$f_{41} = \rho_1^2f_{21}, \quad f_{42} = \rho_2^2f_{22}, \quad f_{43} = L_2C_2s_{11}+RC_2s_{12}+s_{13}$$
$$f_{44} = -L_2C_2s_{16}-RC_2s_{17}-s_{18}, \quad f_{51} = \rho_1^2, \quad f_{52} = \rho_2^2, \quad f_{53} = s_{13}, \quad f_{54} = -s_{18}$$
$$f_{61} = f_{21}/\rho_1, \quad f_{62} = f_{22}/\rho_2, \quad f_{63} = L_2C_2s_{13}+RC_2s_{14}+s_{15}$$
$$f_{64} = -L_2C_2s_{18}-RC_2s_{19}-s_{10}, \quad g_{21} = f_{21}, \quad g_{22} = f_{22}$$
$$g_{23} = L_2C_2s_{21}+RC_2s_{22}+s_{23}, \quad g_{24} = -L_2C_2s_{25}-RC_2s_{26}-s_{27}$$
$$g_{31} = f_{31}, \quad g_{32} = f_{32}, \quad g_{33} = RC_2s_{22}+s_{23}, \quad g_{34} = -RC_2s_{26}-s_{27}$$
$$g_{41} = -\rho_1^3L_1L_2C_2-\rho_1^2C_2\big(L_1(R+R_0)+L_2R_0\big)-\rho_1(L_1+RC_2R_0)-R_0$$
$$g_{42} = -\rho_2^3L_1L_2C_2-\rho_2^2C_2\big(L_1(R+R_0)+L_2R_0\big)-\rho_2(L_1+RC_2R_0)-R_0$$
$$g_{43} = s_{21}L_1L_2C_2-s_{22}C_2\big(L_1(R+R_0)+L_2R_0\big)-s_{23}(L_1+RC_2R_0)-s_{24}R_0$$
$$g_{44} = s_{25}L_1L_2C_2+s_{26}C_2\big(L_1(R+R_0)+L_2R_0\big)+s_{27}(L_1+RC_2R_0)+s_{28}R_0$$
$$g_{51} = \rho_1^2, \quad g_{52} = \rho_2^2, \quad g_{53} = s_{22}, \quad g_{54} = -s_{26}, \quad g_{61} = f_{61}, \quad g_{62} = f_{62}$$
$$g_{63} = L_2C_2s_{22}+RC_2s_{23}+s_{24}, \quad g_{64} = -L_2C_2s_{26}-RC_2s_{27}-s_{28}, \quad x_{21} = \rho_1$$
$$x_{22} = \rho_2, \quad x_{23} = s_{23}, \quad x_{24} = -s_{27}, \quad x_{31} = \rho_1\big(\rho_1^2C_1L_1L_2R_0+\rho_1L_1L_2+R_0(L_1+L_2)\big)$$
$$x_{32} = \rho_2\big(\rho_2^2C_1L_1L_2R_0+\rho_2L_1L_2+R_0(L_1+L_2)\big)$$
$$x_{33} = s_{21}C_1L_1L_2R_0+s_{22}L_1L_2+s_{23}R_0(L_1+L_2)$$
$$x_{34} = -s_{25}C_1L_1L_2R_0-s_{26}L_1L_2-s_{27}R_0(L_1+L_2), \quad x_{51} = -\rho_1^2C_1L_1R_0-\rho_1L_1-R_0$$
$$x_{52} = -\rho_2^2C_1L_1R_0-\rho_2L_1-R_0, \quad x_{53} = -s_{22}C_1L_1R_0-s_{23}L_1-s_{24}R_0$$
$$x_{54} = s_{26}C_1L_1R_0+s_{27}L_1+s_{28}R_0, \quad x_{61} = x_{62} = 1, \quad x_{63} = s_{24}, \quad x_{64} = -s_{28}$$
$$w_{41} = \rho_1^2, \quad w_{42} = \rho_2^2, \quad w_{43} = s_{22}, \quad w_{44} = -s_{26}, \quad w_{2i} = x_{2i}, \quad w_{3i} = x_{3i}$$
$$w_{5i} = x_{5i}, \quad w_{6i} = x_{6i}, \quad u_{2i} = g_{2i}, \quad u_{3i} = g_{3i}, \quad u_{4i} = g_{4i}, \quad u_{6i} = g_{6i}$$

where $i = 1 \cdots 4$ and $k_1 = 1/(\rho_1\rho_2(a^2+b^2))$ and

$$s_{11} = -2\alpha\beta^2(\rho_1+\rho_2)(\alpha^2+\beta^2)^2+\rho_1\rho_2\beta^2(2\alpha^2\beta^2+3\alpha^4-\beta^4)+3\alpha^2\beta^4(\alpha^2+\beta^2)+\beta^2(\alpha^6+\beta^6)$$
$$s_{12} = -\beta^2(\rho_1+\rho_2)(\alpha^2+\beta^2)^2+2\rho_1\rho_2\alpha\beta^2(\alpha^2+\beta^2), \quad s_{13} = \rho_1\rho_2\beta^2(\alpha^2+\beta^2)-\beta^2(\alpha^2+\beta^2)^2$$
$$s_{14} = \beta^2(\rho_1+\rho_2)(\alpha^2+\beta^2)-2\alpha\beta^2(\alpha^2+\beta^2), \quad s_{15} = 2\alpha\beta^2(\rho_1+\rho_2)-\beta^2\rho_1\rho_2+\beta^2(\beta^2-3\alpha^2)$$
$$s_{16} = -\beta(\rho_1+\rho_2)(\beta^4-\alpha^4)(\alpha^2+\beta^2)+\rho_1\rho_2\alpha\beta(2\alpha^2\beta^2-\alpha^4+3\beta^4)-3\alpha^3\beta^3(\alpha^2+\beta^2)-\alpha\beta(\alpha^6+\beta^6)$$
$$s_{17} = \alpha\beta(\alpha^2+\beta^2)^2(\rho_1+\rho_2)+\rho_1\rho_2\beta(\beta^4-\alpha^4)-3\alpha^2\beta^3(\alpha^2+\beta^2)-\beta(\alpha^6+\beta^6)$$
$$s_{18} = \beta(\rho_1+\rho_2)(\alpha^2+\beta^2)^2-\rho_1\rho_2\alpha\beta(\alpha^2+\beta^2)-\alpha\beta(\alpha^2+\beta^2)^2$$

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$$s_{19}=\alpha\beta(\rho_1+\rho_2)(\alpha^2+\beta^2)-\beta\rho_1\rho_2(\alpha^2+\beta^2)+\beta(\beta^4-\alpha^4)$$
$$s_{10}=\beta(\rho_1+\rho_2)(\alpha^2-\beta^2)-\rho_1\rho_2\alpha\beta+(3\beta^2-\alpha^2)\alpha\beta$$
$$s_{21}=-2\alpha\beta^2(\alpha^2+\beta^2)(\rho_1+\rho_2)+\rho_1\rho_2\beta^2(3\alpha^2-\beta^2)+\beta^2(\alpha^2+\beta^2)^2$$
$$s_{22}=-\beta^2(\alpha^2+\beta^2)(\rho_1+\rho_2)+2\alpha\beta^2\rho_1\rho_2,\quad s_{23}=\rho_1\rho_2\beta^2-\beta^2(\alpha^2+\beta^2)$$
$$s_{24}=(\rho_1+\rho_2)\beta^2-2\alpha\beta^2,\quad s_{25}=-\beta(\beta^4-\alpha^4)(\rho_1+\rho_2)+\rho_1\rho_2\alpha\beta(3\beta^2-\alpha^2)-\alpha\beta(\alpha^2+\beta^2)^2$$
$$s_{26}=\alpha\beta(\rho_1+\rho_2)(\alpha^2+\beta^2)-\beta\rho_1\rho_2(\alpha^2-\beta^2)-\beta(\alpha^2+\beta^2)^2$$
$$s_{27}=\beta(\rho_1+\rho_2)(\alpha^2+\beta^2)-\rho_1\rho_2\alpha\beta-\alpha\beta(\alpha^2+\beta^2),\quad s_{28}=\alpha\beta(\rho_1+\rho_2)-\beta\rho_1\rho_2-\beta(\alpha^2-\beta^2).$$

ii) *When The Switch is* OFF

The denominator polynomial of (4) is expressed as

$$\boldsymbol{D}'B=s^4+\boldsymbol{A}'s^3+B's^2+\Gamma's+\Delta' \tag{A.4}$$

where

$$\boldsymbol{A}'=R/L_2,\ B'=1/(L_1C_1)+1/(L_2C_1)+1/(L_2C_2)$$
$$\Gamma'=R/(C_1L_1L_2),\ \Delta'=1/(L_1L_2C_1C_2). \tag{A.5}$$

Equation (6) is expressed by

$$\left.\begin{aligned} i_2'(t)&=G_2'(i_1'(0)+i_2'(0))+G_3'i_3'(0)+G_4'v_c'(0)+G_5'v_r'(0)+G_6'V_{cc}\\ i_3'(t)&=X_2'(i_1'(0)+i_2'(0))+X_3'i_3'(0)+X_4'v_c'(0)+X_5'v_r'(0)+X_6'V_{cc}\\ v_c'(t)&=U_2'(i_1'(0)+i_2'(0))+U_3'i_3'(0)+U_4'v_c'(0)+U_5'v_r'(0)+U_6'V_{cc}\\ v_r'(t)&=W_2'(i_1'(0)+i_2'(0))+W_3'i_3'(0)+W_4'v_c'(0)+W_5'v_r'(0)+W_6'V_{cc}\end{aligned}\right\} \tag{A.6}$$

where

$$G_2'=\mathcal{g}_2'/(C_2L_2),\quad G_3'=\mathcal{g}_3'/(C_2L_2),\quad G_4'=\mathcal{g}_4'/(L_1L_2C_2),\quad G_5'=\mathcal{g}_5'/L_2$$
$$G_6'=\mathcal{g}_6'/(L_1C_2L_2),\quad X_2'=x_2'/(C_1L_2),\quad X_3'=x_3'/(C_1L_1L_2),\quad X_4'=G_5'$$
$$X_5'=x_5'/(C_1L_1L_2),\quad X_6'=x_6'/(C_1L_1L_2),\quad W_2'=\omega_2'/(C_1C_2L_2)$$
$$W_3'=\omega_3'/(C_1C_2L_1L_2),\quad W_4'=\omega_4'/(C_2L_2),\quad W_5'=1+\omega_5'/(C_1C_2L_1L_2)$$
$$W_6'=w_6'/(C_1C_2L_1L_2),\quad U_2'=\omega_2'/(C_1C_2L_2),\quad U_3'=\omega_3'/(C_1C_2L_2)$$
$$U_4'=1+\omega_4'/(C_1C_2L_1L_2),\quad U_5'=W_4'C_2/C_1,\quad U_6'=\omega_6'/(C_1C_2L_1L_2)$$

where for $j=2\cdots6$

$$\mathcal{g}_j'=g_{j1}\sigma_a+g_{j2}\sigma_b+g_{j3}\sigma_c+g_{j4}\sigma_d,\quad x_j'=x_{j1}\sigma_a+x_{j2}\sigma_b+x_{j3}\sigma_c+x_{j4}\sigma_d$$
$$\omega_j'=w_{j1}r_a+w_{j2}r_b+w_{j3}r_c+w_{j4}r_d,\quad \omega_j'=u_{j1}r_a+u_{j2}r_b+u_{j3}r_c+u_{j4}r_d.$$

The coefficients $\sigma_{a\cdots d}$, $r_{a\cdots d}$, f_{ji}, g_{ji}, x_{ji}, w_{ji}, and u_{ji} are functions of the circuit element values and of the roots of the polynomial given in (A.4). When the roots are all complex ($\alpha'\pm j\beta', \omega'\pm jz'$).

$$\sigma_a=e^{x't}\cos(y't)/(s_1^2+s_2^2),\quad \sigma_b=e^{x't}\sin(y't)/(s_1^2+s_2^2)$$
$$\sigma_c=e^{\omega't}\cos(z't)/(s_3^2+s_4^2),\quad \sigma_d=e^{\omega't}\sin(z't)/(s_3^2+s_4^2)$$
$$r_a=\frac{e^{x't}(x'\cos(y't)+y'\sin(y't))-x'}{(s_1^2+s_2^2)(x'^2+y'^2)},\quad r_b=\frac{e^{x't}(x'\sin(y't)-y'\cos(y't))+y')}{(s_1^2+s_2^2)(x'^2+y'^2)}$$
$$r_c=\frac{e^{\omega't}(\omega'\cos(z't)+z'\sin(z't))-\omega'}{(s_3^2+s_4^2)(\omega'^2+z'^2)},\quad r_d=\frac{e^{\omega't}(\omega'\sin(z't)-z'\cos(z't))+z'}{(s_3^2+s_4^2)(\omega'^2+z'^2)}$$

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where

$$s_1 = 2y'^2(\omega' - x'),\ s_2 = y'(x'^2 + \omega'^2 + z'^2 - 2\omega'x' - y'^2)$$
$$s_3 = 2z'^2(x' - \omega'),\ s_4 = z'(\omega'^2 + x'^2 + y'^2 - 2\omega'x' - z'^2)$$
$$s_{1a} = (x'(x'^2 - 3y'^2))s_1 + (y'(3x'^2 - y'^2))s_2,\ s_{2a} = (x'^2 - y'^2)s_1 + 2x'y's_2$$
$$s_{3a} = x's_1 + y's_2,\ s_{4a} = s_1,\ s_{5a} = (y'(3x'^2 - y'^2))s_1 - (x'(x'^2 - 3y'^2))s_2$$
$$s_{6a} = 2x'y's_1 - (x'^2 - y'^2)s_2,\quad s_{7a} = y's_1 - x's_2,\quad s_{8a} = -s_2$$
$$s_{1b} = (\omega'(\omega'^2 - 3z'^2))s_3 + (z'(3\omega'^2 - z'^2))s_4,\quad s_{2b} = (\omega'^2 - z'^2)s_3 + 2\omega'z's_4$$
$$s_{3b} = \omega's_3 + z's_4,\quad s_{4b} = s_3,\quad s_{5b} = (z'(3\omega'^2 - z'^2))s_3 - (\omega'(\omega'^2 - 3z'^2))s_4$$
$$s_{6b} = 2\omega'z's_3 - (\omega'^2 - z'^2)s_4,\quad s_{7b} = z's_3 - \omega's_4,\quad s_{8b} = -s_4$$
$$g_{21} = L_2C_2s_{1a} + RC_2s_{2a} + s_{3a},\quad g_{22} = -L_2C_2s_{5a} - RC_2s_{6a} - s_{7a}$$
$$g_{23} = L_2C_2s_{1b} + RC_2s_{2b} + s_{3b},\quad g_{24} = -L_2C_2s_{5b} - RC_2s_{6b} - s_{7b},\quad g_{31} = RC_2s_{2a} + s_{3a}$$
$$g_{32} = -RC_2s_{6a} - s_{7a},\quad g_{33} = RC_2s_{2b} + s_{3b},\quad g_{34} = -RC_2s_{6b} - s_{7b}$$
$$g_{41} = -C_2(L_1 + L_2)s_{2a} - RC_2s_{3a} - s_{4a},\quad g_{42} = C_2(L_1 + L_2)s_{6a} + RC_2s_{7a} + s_{8a}$$
$$g_{43} = -C_2(L_1 + L_2)s_{2b} - RC_2s_{3b} - s_{4b},\quad g_{44} = C_2(L_1 + L_2)s_{6b} + RC_2s_{7b} + s_{8b}$$
$$g_{51} = s_{2a},\quad g_{52} = -s_{6a},\quad g_{53} = s_{2b},\quad g_{54} = -s_{6b},\quad g_{61} = L_2C_2s_{2a} + RC_2s_{3a} + s_{4a}$$
$$g_{62} = -L_2C_2s_{6a} - RC_2s_{7a} - s_{8a},\quad g_{63} = L_2C_2s_{2b} + RC_2s_{3b} + s_{4b}$$
$$g_{64} = -L_2C_2s_{6b} - RC_2s_{7b} - s_{8b},\quad x_{21} = s_{3a},\quad x_{22} = -s_{7a},\quad x_{23} = s_{3b},\quad x_{24} = -s_{7b}$$
$$x_{31} = L_1L_2C_1s_{1a} + (L_1 + L_2)s_{3a},\quad x_{32} = -L_1L_2C_1s_{5a} - (L_1 + L_2)s_{7a}$$
$$x_{33} = L_1L_2C_1s_{1b} + (L_1 + L_2)s_{3b},\quad x_{34} = -L_1L_2C_1s_{5b} - (L_1 + L_2)s_{7b}$$
$$x_{51} = -L_1C_1s_{2a} - s_{4a},\quad x_{52} = L_1C_1s_{6a} + s_{8a},\quad x_{53} = -L_1C_1s_{2b} - s_{4b}$$
$$x_{54} = L_1C_1s_{6b} + s_{8b},\quad x_{61} = s_{4a},\quad x_{62} = -s_{8a},\quad x_{63} = s_{4b},\quad x_{64} = -s_{8b}.$$

For $i = 1 \cdots 4$,

$$w_{2i} = x_{2i},\quad w_{3i} = x_{3i},\quad w_{4i} = g_{5i},\quad w_{5i} = x_{5i},\quad w_{6i} = x_{6i},\quad u_{2i} = g_{2i}$$
$$u_{3i} = g_{3i},\quad u_{4i} = g_{4i},\quad u_{5i} = g_{5i},\quad u_{6i} = g_{6i}.$$

iii) Finally the elements of the matrices $\mathbf{W}$ and $\mathbf{Z}$ of (9), are calculated for $t = t_1$ and $t = t_2$, i.e., the instants at which the switch turns OFF and ON, respectively, from the expressions

$$W_{1j_{*1}} = G_2'(F_{j+1} + G_{j+1}) + G_3'X_{j+1} + G_4'U_{j+1} + G_5'W_{j+1},\quad W_{11} = W_{1j_{*1}} - 1$$
$$W_{2j_{*2}} = X_2'(F_{j+1} + G_{j+1}) + X_3'X_{j+1} + X_4'U_{j+1} + X_5'W_{j+1},\quad W_{22} = W_{2j_{*2}} - 1$$
$$W_{3j_{*3}} = U_2'(F_{j+1} + G_{j+1}) + U_3'X_{j+1} + U'_4U_{j+1} + U_5'W_{j+1},\quad W_{33} = W_{3j_{*3}} - 1$$
$$W_{4j_{*4}} = W_2'(F_{j+1} + G_{j+1}) + W_3'X_{j+1} + W_4'U_{j+1} + W_5'W_{j+1},\quad W_{44} = W_{4j_{*4}} - 1$$
$$Z_1 = -G_2'(G_6 + F_6) - G_3'X_6 - G_4'U_6 - G_5'W_6 - G_6',\quad Z_2 = -X_2'(G_6 + F_6) - X_3'X_6 - X_4'X_6 - X_5'W_6 - X_6'$$
$$Z_3 = -U_2'(G_6 + F_6) - U_3'X_6 - U_4'U_6 - U_5'W_6 - U_6',\quad Z_4 = -W_2'(G_6 + F_6) - W_3'X_6 - W_4'U_6 - W_5'W_6 - W_6'$$

with

$$\mathbf{X}^T(0) = [i_1(0) + i_2(0)\quad i_3(0)\quad v_c(0)\quad v_r(0)].$$

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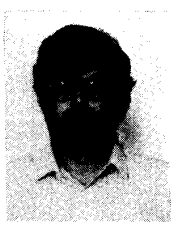
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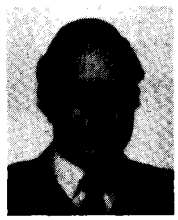
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